

Limited Conservation Benefits From the First Marine Protected Area in the High Seas

Juan Carlos Villaseñor-Derbez^{1,2} and John Lynham³

¹Rosenstiel School of Marine, Atmospheric & Earth Sciences, University of Miami

²Institute for Data Science & Computing, University of Miami

³Department of Economics, University of Hawai'i at Manoa

This version: July 31, 2026

Abstract

Can Large-Scale Marine Protected Areas be effective on the high seas? We address this question by evaluating the world's first large-scale spatial closure implemented in areas beyond national jurisdiction. In the Western and Central Pacific Ocean, two large high-seas pockets were closed to industrial purse-seine fishing in 2010 to reduce bycatch of juvenile bigeye tuna, but industrial longline fishing was allowed to continue. Nearly two decades later, the empirical consequences of this policy change remain unknown. We use publicly available data to evaluate this policy and find that the closure significantly reduced purse-seine fishing within the high-seas pockets and that longline vessels operating within the closed areas experienced a weakly significant 19.6% increase in the number of bigeye tuna caught per 1000 hooks. We find no evidence that the closure generated a spillover benefit to purse-seine vessels fishing near the boundaries of the closed areas.

Significance statement. The High Seas Treaty entered into force in January 2026, yet no empirical evidence exists on whether Marine Protected Areas can deliver conservation benefits beyond national jurisdiction. We provide this evidence by evaluating the world's first large-scale high-seas closure—two vast Pacific fishing grounds totaling over one million km² closed to industrial purse seining since 2010. Analyzing two decades of spatially explicit fishing data, we show that market-based enforcement successfully eliminated industrial purse seine fishing from a large swath of international waters. However, the resulting conservation benefits were far more modest than those documented for comparable closures within national waters, and no spillover reached adjacent fisheries. As nations begin designating high-seas MPAs under the new treaty, our findings provide the only empirical benchmark for what these protections may—and may not—achieve.

1 Introduction

The entry into force of the UN High Seas Treaty (BBNJ Agreement) on January 17, 2026, officially established the legal framework necessary to create and manage Marine Protected Areas (MPAs) on the high seas. By definition, high seas MPAs will protect pelagic ecosystems. In this way, they will share some similarities with Large-Scale MPAs (LSMPAs; areas larger than 100,000 km² [1, 2]). By virtue of the habitats they protect, LSMPAs most often restrict fishing effort of pelagic fisheries, such as those targeting tuna, billfish, and sharks [3]. This is also likely to be one of the primary impacts of high seas MPAs.

Previous work has found that tuna fisheries operating near LSMPA boundaries may see increases in their catch rates [4, 5, 6]. For example, Hilborn et al. [6] finds that the CPUE of bigeye tuna has increased by 13% outside the boundaries of Papahānaumokuākea Marine National Monument in Hawai'i. This spillover benefit may prompt nations to think strategically when designing LSMPAs on the high seas: nations may implement LSMPAs immediately outside their Exclusive Economic Zones (EEZs) hoping to boost catch rates in their waters [7, 8]. But would this work?

There are subtle but important differences between Large-Scale MPAs inside EEZs and Large-Scale MPAs on the high seas. One important difference is enforcement and compliance. Not all nations have ratified the UN Convention on the Law of the Sea and it is still unclear who would be responsible for enforcing and potentially confronting illegal fishing in future high seas MPAs. A second important difference is habitat and environmental conditions. High seas MPAs will all be located at least 200 nautical miles away from land, in some of the deepest and most remote marine ecosystems in the world. Many of the processes that influence pelagic species around islands and seamounts are either absent or very different on the high seas [9, 10, 11, 12]. Therefore, results from LSMPAs within national jurisdictions may not extrapolate to similarly sized MPAs on the high seas [13]. Here, we fill this crucial gap in policy-relevant knowledge by studying the first known case of large-scale area-based marine conservation on the high seas.

We focus on the Western and Central Pacific Ocean, where industrial tuna fishing vessels use two main gears to target different tuna species. The longline fleet primarily catches albacore (*Thunnus alalunga*), bigeye (*Thunnus obesus*), and yellowfin tuna (*Thunnus albacares*). The purse seine fleet mainly catches skipjack (*Katsuwonus pelamis*) and yellowfin tuna, with some bycatch of juvenile bigeye tuna associated with the use of drifting Fish Aggregating Devices (dFADs) [14, 15, 16].

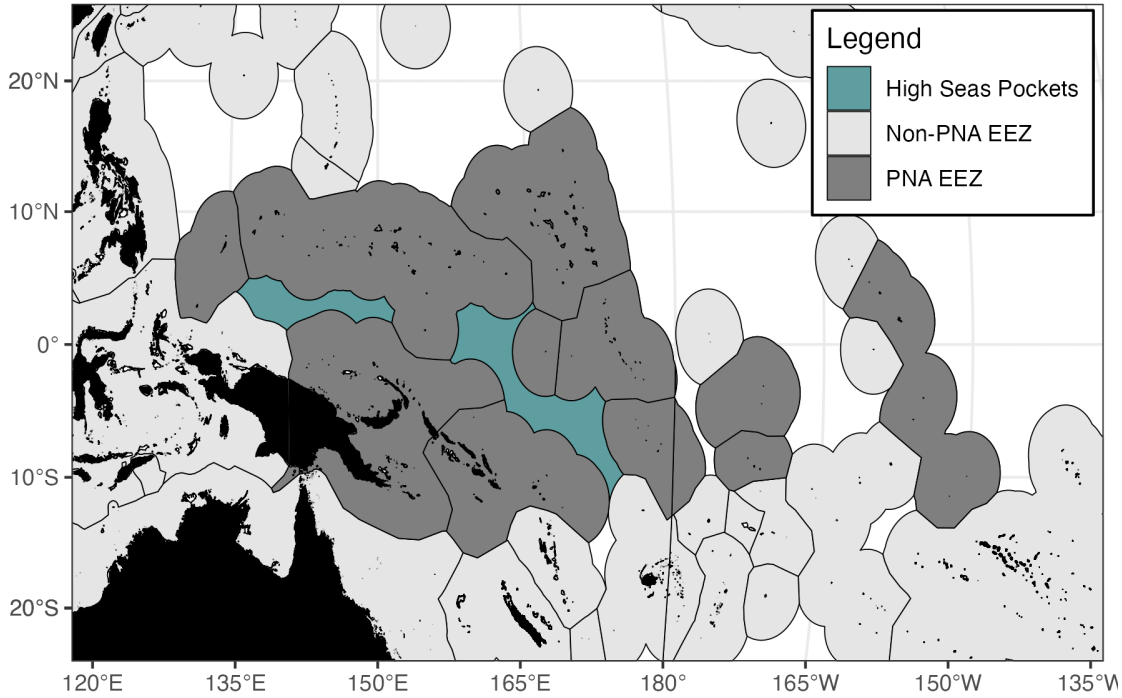


Figure 1: Map of the Western and Central Pacific Ocean. Light gray polygons show the Exclusive Economic Zones of non-PNA nations. Dark gray polygons highlight the Exclusive Economic Zones of the eight Parties to the Nauru Agreement nations and Tokelau. The two teal polygons show the High Seas Pockets—the “donut holes” formed by the surrounding Exclusive Economic Zones—where purse seine fishing has been prohibited since 2010.

The Parties to the Nauru Agreement (PNA¹)—a group of eight nations in the region—and Tokelau jointly manage tuna fisheries in their waters using a market-based approach known as the Vessel Day Scheme (VDS; [17, 18, 19]). Bycatch of juvenile tuna by the

¹PNA members include: Federated States of Micronesia, Kiribati, Nauru, Palau, Papua New Guinea, Republic of the Marshall Islands, Solomon Islands, and Tuvalu.

67 purse seine fleet presented a major problem to the longline fleet [14, 15, 16], particularly
 68 in the two high seas pockets formed between the PNA’s Exclusive Economic Zones (EEZs;
 69 Figure 1). When the spatial overlap between target and bycatch species is severe, MPAs
 70 may be a useful tool to protect the “weak stock” [20], here juvenile bigeye. In an effort
 71 to reduce the purse-seine fleet’s bycatch of juvenile bigeye tuna—an externality affecting
 72 longline vessels targeting adult bigeye tuna—purse seine fishing was prohibited in the high
 73 seas pockets starting in 2010 [21, 22]. These pockets are roughly 360,000 km² and 780,000
 74 km², so they are very much in line with the size of LSMPAs in the region, like the Palau
 75 National Marine Sanctuary (475,077 km²) or the now degazetted Phoenix Islands Protected
 76 Area (408,422 km²). In closing the high seas pockets to industrial tuna purse seine fisheries,
 77 this intervention effectively created the first partially protected *de facto* large-scale MPA in
 78 areas beyond national jurisdiction. How was the prohibition on purse seine fishing achieved
 79 before the UN High Seas Treaty even existed? The PNA used a carrot-and-stick approach:
 80 vessels that adhered to the closure were allowed to fish in the lucrative waters of the PNA
 81 countries under the VDS program; vessels that ignored the purse seine ban were excluded
 82 from the VDS program².

83 There are only a few examples of high seas closures implemented before the adoption of
 84 the high seas treaty, and their potential effectiveness remains contested or unevaluated [13].
 85 Previous work has claimed that the benefits of high seas closures to tuna populations would
 86 be limited. For example, Sibert et al. [34] used hind-cast simulations to estimate the impact
 87 that the PNA’s high seas pocket closures would have on bigeye and skipjack tuna populations,
 88 finding that they would have negligible stock-wide effects (0.1-0.8% increase) but potential
 89 for spillover (10-25% localized increase), depending on model assumptions. However, nearly
 90 two decades later, the actual effectiveness of this spatial intervention remains untested. More
 91 importantly, recent empirical work on LSMPAs within national jurisdictions has found that

²The WCPFC introduced a small exception for the Philippine artisanal purse seine fleet, which allowed limited fishing by a small number of vessels in the north-western pocket, as stated in a series of conservation and management measures [23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33]. We address this with robustness tests that separately evaluate each pocket in Supplementary Materials section C.1.

catch rates tend to increase in areas near LSMPA boundaries and abundances may increase locally [4, 5, 6].

Our objective is to test for the effectiveness of this unique policy intervention: the first large-scale spatial closure on the high seas. We compile a multi-decadal (2000-2020) and spatially explicit dataset on catch and effort by purse seine and longline fisheries using publicly available information from the Western and Central Pacific Fisheries Commission (WCPFC; see section 3). We present our analysis in three parts. First, we ask whether the intervention actually reduced purse seine fishing in the high seas pockets. Second, we test whether the reduction of purse seine fishing effort benefited the longline fleet that still fishes in the pockets (as originally hoped for). Third, we test whether the high seas pockets, acting as a *de facto* marine protected area, provided a spillover benefit to tuna fisheries fishing outside but near the high seas boundaries (an unintended benefit observed in other settings). We answer these questions following a “pre-registered” study design available on the Open Science Framework platform [35]. As we will show, the intervention was successful at eliminating purse seine fishing from the high seas pockets, but we only observe weak evidence that this led to a benefit to the longline fleet, and no evidence of a spillover benefit. We believe our findings can help inform the design of, and manage expectations around, proposed LSMPAs on the high seas.

2 Results and Discussion

2.1 Reducing fishing effort by purse seiners

The closures sought to eliminate purse seine fishing effort from two high seas pockets. Consequently, we begin by testing for changes in purse seine fishing effort in the high seas pockets. A visual inspection of trends in the data shows that, in the 10 years leading to the closure, total annual fishing effort by purse seiners fishing in the high seas pockets was around $2,187 \pm 770$ days (mean $\pm$ standard deviation) or $1,766 \pm 634$ sets (Figure 2A-B). After the clo-

sure, fishing effort in the pockets dropped to zero starting in 2010, except for a few events recorded in 2013 and 2019.

To conclusively claim that the reduction in fishing effort is due to the policy change, we must compare trends in fishing effort inside the high seas pockets to trends in fishing effort in nearby comparable areas. These “control” areas must be subject to the same environmental conditions as well as fishery management rules, thus providing a counterfactual of expected fishing effort if the closure had not been implemented. With this in mind, we compare all $1^\circ \times 1^\circ$ grid cells in the high seas pockets ($N = 40$) to other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S ($N = 151$; see Figure A3). We estimate a two-way fixed effects regression model that accounts for differences across grid cells and years to test for a causal effect of the closure (see Methods in section 3.4.1 for more details).

In line with the patterns in Figure 2, we find that the closure dramatically reduced fishing effort in the high seas pockets, with an estimated annual reduction of 59.47 days or 50.93 sets per grid cell per year ($p < 0.01$), relative to trends observed for other high-seas areas in the region (Table 1). We also estimate a model with dynamic treatment effects that measures the change in fishing effort by year. These event-study regressions reveal two important findings. First, that pre-treatment estimates are small relative to the post-treatment effect and show no systematic trend, though a few individual years differ from zero. And second, that all years since 2010 had significantly less fishing effort than what would have been expected, had the closure never been implemented ($p < 0.01$; Figure 2C-D). Results for other specifications are provided in Supplementary Materials section B.1.

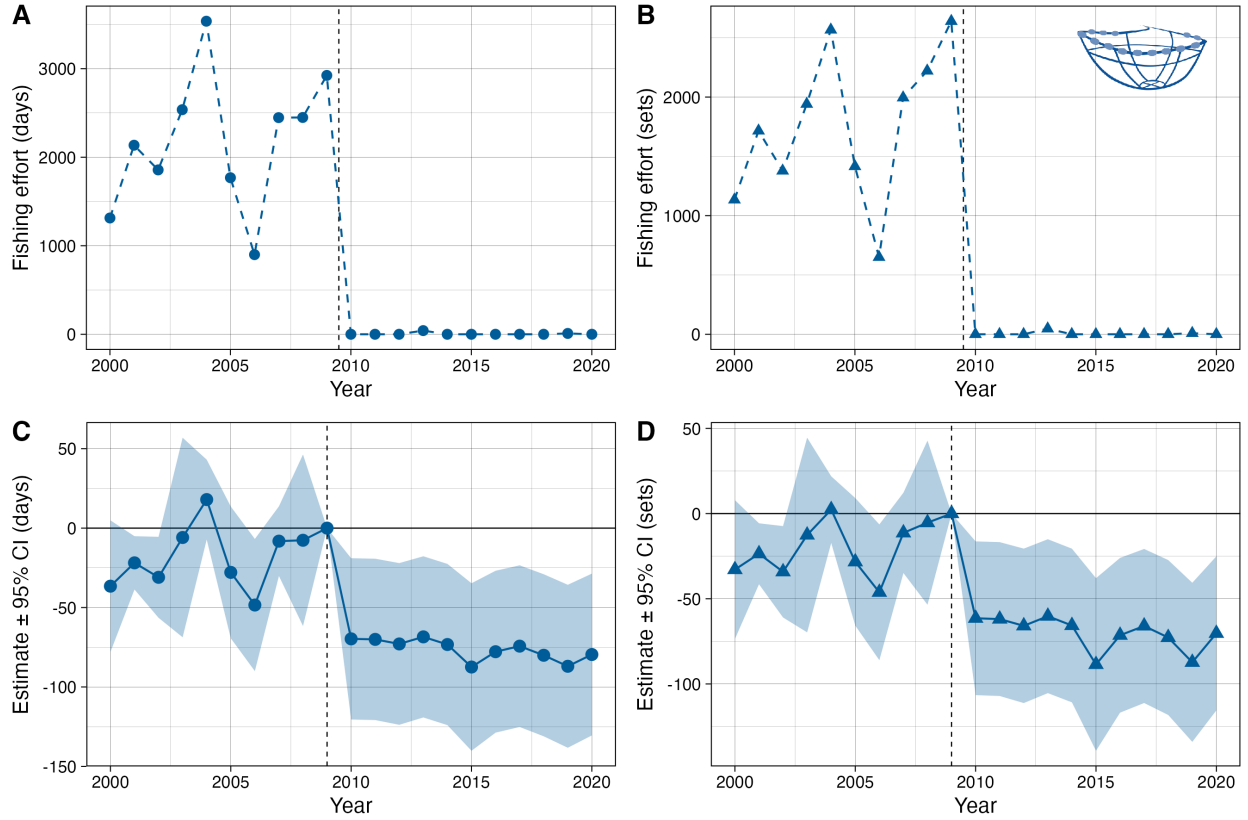


Figure 2: Change in purse seine fishing effort in the high seas pockets (combined). Panels A and B show annual total fishing effort measured in days and number of sets, respectively, across all grid cells in the high seas pockets. Panels C and D show event-study plots for the change in fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area.

Table 1: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure, relative to changes in fishing effort observed for other comparable high seas areas in the WCPFC convention area.

	Effort (days)	Effort (sets)
Post x Treated	−59.470*** (13.762)	−50.933*** (10.607)
$\bar{Y}_{pre}$	54.670	44.145
Num.Obs.	4011	4011
R2 Adj.	0.388	0.301

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

2.2 Benefits to the longline fishery

In the 10 years before closure (2000-2009), tuna purse seines caught an average of 3,200 metric tons (mt) of bigeye tuna³, 48,500 mt of skipjack tuna, and 13,800 mt of yellowfin tuna per year inside the two high seas pockets (Figure A1). Our results show that purse seine fishing effort was significantly reduced in the high seas pockets in 2010. It follows, then, that so was catch by purse seiners. Recall that the longline fleet has always been allowed to operate in the high seas pockets, so now we ask: Did reducing purse seine fishing mortality in the high seas pockets benefit the longline fleet?

We focus on bigeye tuna because the closure was motivated by high bycatch rates of juvenile bigeye in the purse seine fishery [14, 34, 15, 16]. We measure benefits to the longline fleet using catch-per-unit-effort (CPUE) of bigeye tuna. Specifically, we measure CPUE in number of fish for every thousand hooks and metric tons for every thousand hooks, which

³To put 3,200 metric tons in context, the Total Allowable Catch for the US-flagged longline fleet for the entire Western and Central Pacific in 2024 was 3,554 metric tons.

can respectively be understood as imperfect proxies for local abundance and biomass [36, 37]. Time series of both measures of bigeye CPUE from the longline fleet operating in the high seas pockets and control regions are shown in Figure 3A-B. Both measures show a steady decline in CPUE during the ten years before the exclusion of purse seiners. After the closure, the decreasing trend slows down and CPUE shows relative stability, particularly when measured in number of fish per thousand hooks (Figure 3A). Note, too, that there was a sharp increase in CPUE starting around three years after the closure, peaking at around five years, and then returning to a steady level seven years after the closure.

As before, a formal test requires that we compare trends in CPUE in the high seas pockets to an appropriate control region. In this second test, we retain all $5^\circ \times 5^\circ$ grid cells⁴ that fall in the high seas pockets (our “treated” cells; $N = 16$) and all high seas grid cells in the WCPFC convention area that lie between 20°N and 20°S ($N = 69$; Figure A4). We then estimate annual changes in bigeye CPUE in high seas pocket grid cells, relative to changes observed in our counterfactual grid cells (see Methods section 3.4.2). We detect a modest increase in bigeye CPUE of 0.70 fish per thousand hooks ($p < 0.1$), and no change in CPUE when measured in tons per thousand hooks (Table 2). In terms of percentage changes, the increase in bigeye CPUE represents a 19.6% increase above the pre-treatment mean. The contrasting results across the two measures of CPUE suggest that fishers may have caught more fish per unit effort between 2010 and 2020, but that these fish were, on average, smaller. It is also very likely that the weight measure has more measurement error (it is easy, and standard practice, to record every bigeye caught, but not every fish is weighed individually, and weights may be approximations made by the crew or extrapolated at the point of landing). Measurement error will bias coefficient estimates towards zero, in line with what we observe here.

Our event-study plots echo these aggregate results: the annual number of bigeye tuna caught per thousand hooks exhibits a slightly positive trend since 2010, but we observe no

⁴Longline data are reported on a $5^\circ \times 5^\circ$ grid, while purse seine data are reported using a $1^\circ \times 1^\circ$ grid.

change in bigeye CPUE when measured in metric tons per thousand hooks (Figure 3C-D). The lack of a monotonic increase in bigeye CPUE could be caused by a subsequent increase in fishing effort following the localized and temporary recovery of the stock. A similar pattern appears for yellowfin tuna, where we estimate a 39% increase in CPUE measured in numbers of fish per thousand hooks (coefficient = 0.329, $p < 0.05$; Table B6). We read this result as suggestive because it is present only in the log-transformed specification and only in numbers rather than weight. Yellowfin was also not the focal species of our pre-registered hypothesis, and is one of several species-by-specification combinations we examine. Results for other specifications and species are presented in the Supplementary Materials section B.2.

Because the two pockets differ in size, degree of protection, and surrounding habitat, we also estimate our main specifications separately for each pocket. The increase in bigeye CPUE is present in both pockets, but it is nearly four times larger in the smaller, north-western pocket (HSP1) than in the larger, eastern one (HSP2), with coefficient estimates of 1.537 fish per thousand hooks ($p < 0.01$) compared to 0.397 ($p < 0.05$), respectively (Table C1). This runs counter to the intuition that a larger closure should confer larger benefits, though two features of HSP1 caution against reading too much into the contrast. First, the HSP1 estimate rests on only five treated grid cells, compared to eleven in HSP2. Second, HSP1 is the pocket where the Philippines' purse seine fleet retained limited access, making it the less protected of the two.

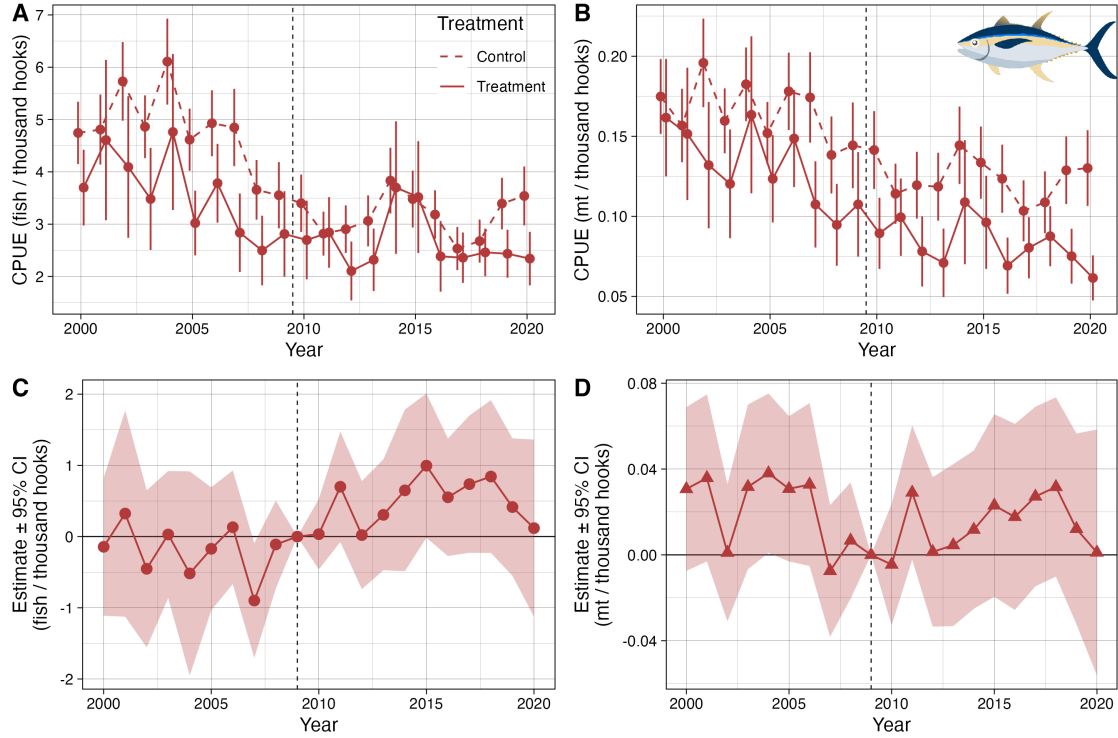


Figure 3: Change in catch-per-unit-effort (CPUE) of bigeye tuna in the longline fleet. Panels A and B show time series of mean CPUE in number of fish per 1000 hooks and metric tons per 1000 hooks, respectively, across all grid cells in the high seas pockets (solid lines) and other high seas grid cells (dashed lines). Panels C and D show event-study plots for the change in bigeye tuna CPUE in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area.

Table 2: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area.

	fish / 1000 hooks	mt / 1000 hooks
Post x Treated	0.697* (0.363)	-0.003 (0.011)
$\bar{Y}_{pre}$	3.548	0.131
Num.Obs.	1631	1631
R2 Adj.	0.798	0.829

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

2.3 Spillover benefits to the purse seine fishery

Skipjack tuna in the open ocean are mainly targeted by the purse seine fleet, with the longline fishery representing less than 1% of total skipjack caught in the WCPFC convention area [38]. When the high seas pockets were closed to purse seine fishing, the region essentially became a *de facto* no-take zone for skipjack tuna (at least for the eastern pocket, HSP2, not subject to the Philippines exemption). Previous work has shown that the CPUE of tuna purse seiners often, but not always, increases near large-scale MPAs after their implementation [5]. Therefore, we now ask whether the high seas pockets, by substantially reducing localized fishing mortality on skipjack, also produced spillover benefits to tuna purse seine fisheries operating near the borders of the pockets. A direct examination of a time series of skipjack tuna CPUE in Figure 4 shows two contrasting trends. When measured in metric tons *per day*, CPUE within 100 nautical miles of the high seas pockets appears to increase after 2010, as compared to pre-closure years (Figure 4A). We observe the opposite when we measure CPUE in metric tons *per set*, which shows lower CPUE for most years after the closure

(Figure 4B).

Following previous work [4, 5, 6], we formally test for changes in skipjack CPUE near the high seas pocket boundaries (within 100 nm; $N = 226$), relative to changes observed far from the boundary (100–200 nm; $N = 184$; Figure A5). We only retain grid cells that fall within PNA countries’ waters because purse seine fishing is strictly managed under the vessel-day scheme in these waters [17, 5]. Using a standard Before-After-Near-Far design, we fail to detect changes in CPUE that would be indicative of a spillover effect, with estimated effects of -0.50 tons per day and -0.87 tons per set ($p = 0.45$ and $p = 0.29$, respectively; Table 3). Our event-study estimates are shown in panels C and D of Figure 4. These highlight that CPUE near the high seas pockets is not different from CPUE observed 100–200 nm from the pockets. Overall, there is no evidence to indicate that closing the high seas pockets produced spillover benefits to adjacent purse seine vessels targeting skipjack tuna. This null result holds when we estimate the model separately for each pocket (Table C2). Notably, we detect no spillover even around the larger eastern pocket (-0.952 tons per day, $p = 0.32$), where the greater area of fully displaced fishing mortality would give the strongest reason to expect one. Similar results for the longline fleet, other species, and additional model specifications are presented in the Supplementary Materials section B.3.

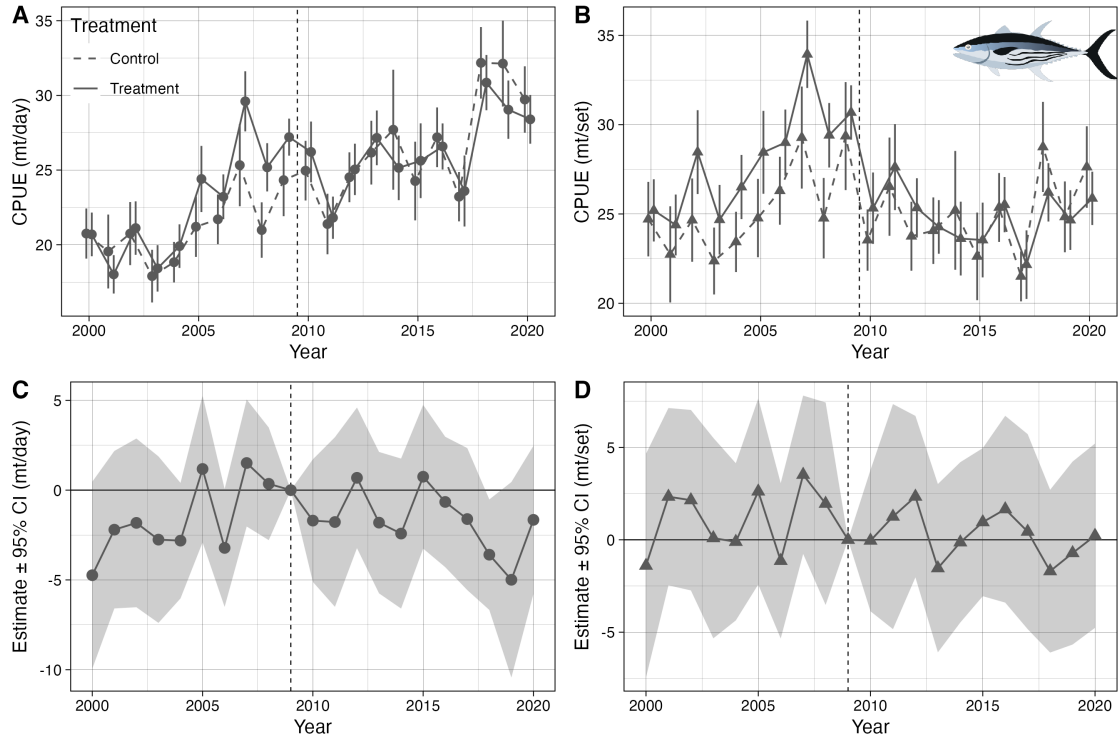


Figure 4: Change in catch-per-unit-effort (CPUE) of skipjack tuna in the purse seine fleet. Panels A and B show time series of mean CPUE in tons per day and tons per set, respectively, across all grid cells within 100 nm of the high seas pockets (solid) and those far away (100–200 nm, dashed). Panels C and D show coefficient estimates for the change in skipjack tuna CPUE near the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near and far purse seine samples are restricted to grid cells within PNA member Exclusive Economic Zones.

Table 3: Coefficient estimates for change in skipjack tuna CPUE in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation’s Exclusive Economic Zones.

	CPUE (mt/day)	CPUE (mt/set)
Post x Near	−0.496 (0.679)	−0.865 (0.818)
$\bar{Y}_{pre}$	22.946	28.182
Num.Obs.	5960	5960
R2 Adj.	0.333	0.245

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius. $\bar{Y}_{pre}$ indicates the mean of each outcome variable over the pre-closure period.

2.4 Policy Implications for Large-Scale MPAs on the High Seas

We evaluated changes in fishing effort and catch-per-unit effort inside and near two high seas areas where a closure on purse seine fishing has been implemented since 2010. Our results show that purse seine fishing effort was significantly curtailed in the two high seas pockets. This intervention led to longliners catching 19.6% more bigeye tuna per unit effort, temporarily. But, we do not observe evidence that the closure boosted catch rates of skipjack tuna for purse seine vessels fishing near the high seas pockets.

The success in eliminating fishing effort within the high seas pockets highlights the power of the environmental market implemented by the Parties to the Nauru Agreement (PNA). PNA waters contain some of the most productive skipjack tuna fishing grounds in the world and the majority of the Pacific resource is typically caught in these waters [17, 18]. The combination of high environmental productivity and strong market power gives PNA members considerable bargaining power when negotiating access agreements for foreign fishing

vessels. As we have shown, this allowed the PNA members to close a portion of the high seas to purse seine fishing, well before the High Seas Treaty was even ratified. This result contributes to the literature on the value of extending property rights over a portion of a common pool resource [39, 40]: by leveraging the demand for resources in their waters, the PNA countries essentially established a property right in an area with no formal property rights.

Even though the intervention successfully reduced purse seine fishing in the high seas pockets, the evidence of benefits to the longline fleet is somewhat limited. We find that longliners catch more fish, but not more tons of fish, per unit effort. This is broadly aligned with initial expectations. Because purse seiners are no longer incidentally catching juvenile bigeye tuna, more of these fish are available for harvest by the longline fleet. It is plausible that there are cohorts of juvenile bigeye that evade purse seine mortality within the pockets and some survivors later show up on longline hooks. But continued or even higher longline fishing mortality within the pockets would reduce the probability of survival into larger size classes.

The weak and null effects we detect could be due to the resolution of the data. Longline data are reported on a $5^\circ \times 5^\circ$ grid, which may be too coarse to detect the expected small effects [34]. Unfortunately, the WCPFC does not make higher resolution data publicly available but, in principle, these exist and would allow for a more granular test as done elsewhere [41, 5, 6]. Of course, another explanation is that the high seas closures are simply not providing any conservation benefits, suggesting that traditional fisheries management approaches such as regulating the use of drifting Fish Aggregating Devices across the region could have been more effective [34, 16].

We fail to detect any benefit to purse seine fisheries operating near the high seas pockets, even though the closure significantly reduced fishing mortality for skipjack tuna. This may be because longliners continued operating in the region and they occasionally catch skipjack tuna. This lingering source of fishing mortality means that skipjack are only partially pro-

267 tected, and previous work has shown that fully protected MPAs are the ones most likely to
268 deliver conservation benefits [42].

269 Differences in habitat are another possible explanation for the null findings compared to
270 MPAs situated inside of EEZs where spillover has been documented [4, 5, 6]. For example,
271 previous work has shown that pelagic species tend to aggregate around shallow (< 150 m
272 deep) seamounts and islands [9, 10, 11, 12, 43]. Most Large-Scale MPAs within EEZs are
273 placed around volcanic islands, so it is not surprising that these MPAs contain multiple is-
274 lands and shallow seamounts that produce the heterogeneous habitat that aggregates highly
275 mobile pelagic species. The most comprehensive global dataset on seamount location and
276 depth [44, 45] shows that the high seas pockets do not contain any shallow seamounts (Fig-
277 ure A2). This observation would lend support to calls for high seas MPAs to focus on these
278 underwater features [46].

279 In conclusion, our results show that PNA members were able to use their bargaining
280 power—rooted in successful fisheries management—to significantly reduce fishing effort in the
281 high seas pockets. This intervention led to the world’s first *de facto* large-scale MPA in areas
282 beyond national jurisdiction. We find mixed evidence around the effects of this significant
283 reduction in fishing effort, indicating that spillover benefits from high seas MPAs may be
284 weaker than those observed to date around similar-sized MPAs within EEZs. Our results also
285 show that lessons learned from LSMPAs implemented within national jurisdictions may not
286 directly translate to those implemented in the high seas. They also highlight how features
287 such as degree of protection, amount of fishing displaced, and presence of critical habitat
288 may drive potential outcomes of future *de jure* MPAs in the high seas.

289 3 Methods

290 Prior to running any empirical analysis, we pre-registered our analysis plans using the As-
291 Predicted template in the Open Science Framework. Pre-registration plans aim to prevent

p-hacking, *ex post* hypothesizing, and discarding important null findings [47, 48]. The pre-registration plan states the hypotheses being tested, the outcome variables and units of measurement, and the process by which conditions (e.g. treated vs. control) are assigned. Our pre-registered plan is available at [35].

We set out to test three hypotheses: (1) the PNA’s closure of the high-seas pockets to purse seine fishing reduced purse seine fishing effort within the high seas pockets; (2) reducing purse seine fishing effort within the high seas pockets will increase catch-per-unit-effort of tuna, especially bigeye tuna, for the longline fleet operating within the high seas pockets; and (3) the closure will also increase catch-per-unit-effort of bigeye, skipjack, and yellowfin tuna for the purse seine and longline fleets operating in PNA waters within 100 nautical miles of the high seas pockets, relative to catch-per-unit-effort 100-200 nautical miles from the high seas pockets. The methods below describe the steps we take to build our panel dataset and test our hypotheses.

3.1 Data sources

We primarily used publicly available data provided by the Western and Central Pacific Fisheries Commission (WCPFC). We downloaded data from the WCPFC Public Domain data portal.⁵ Specifically, the purse seine data came from the following file: “*Aggregated data, grouped by 1°x1° latitude/longitude grids, YEAR and MONTH*” (WCPFC_S_PUBLIC_BY_1x1_MM5). And the longline data came from “*Aggregated data, grouped by 5°x5° latitude/longitude grids, YEAR and MONTH*” (WCPFC_L_PUBLIC_BY_YR_MON_3). Both datasets were downloaded on Oct 31, 2025.

⁵www.wcpfc.int/wcpfc-public-domain-aggregated-catcheffort-data-download-page

3.2 Dataset construction

3.2.1 Outcomes of interest

Our two outcomes of interest are fishing effort and catch-per-unit-effort (CPUE). We measured fishing effort by the purse seine fleet in both days and sets. These are reported monthly as 1° by 1° grid cell totals. We summed the monthly totals to obtain annual measures of effort by grid cell. We calculate CPUE per grid cell and year as the sum of the total catch divided by the total effort reported for that cell, adjusting for gear-specific differences. Longline CPUE is measured in number of fish per thousand hooks and tons per thousand hooks. For purse seine vessels, CPUE is measured in tons of fish per day and tons of fish per set.

3.3 Treatment and Control Groups

All regression-based hypothesis tests include indicator variables for the periods before and after the high seas closures (effective January 2010), as well as for treatment and control groups. The definition of treatment and control groups varies by hypothesis test. When testing for changes in purse seine fishing effort, we define treated grid cells as those within the high seas pockets, and control grid cells as other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S . When testing for changes in CPUE to the longline fleet operating in the high seas pockets, we consider grid cells inside high seas pockets as “treated” and other high seas grid cells that fall under the WCPFC convention area and lie between 20°N and 20°S as “control”. Finally, when testing for spillover benefits, we follow the same Before-After-Near-Far design used to test for spillover benefits from Large-Scale Marine Protected Areas in other settings [4, 5]. Specifically, grid cells within 100 nautical miles of the high seas pockets’ boundaries are labeled as treated. Grid cells between 100 and 200 nautical miles from the boundaries are labeled as controls. For the purse seine fishery, we only retain grid cells that fall within PNA countries’ EEZs to account for the sharp difference in management between vessels that operate under the vessel-day

338 scheme *vs.* those fishing on the high seas [17, 18, 19].

339 3.4 Identification and Estimation

340 3.4.1 Testing for changes in fishing effort inside the high seas pockets

341 Let E_{it} be purse seine effort (measured in sets or days) in pixel i at time t . We first test for
 342 changes in fishing effort in the high seas pockets with a simple linear regression that does
 343 not account for changes in region-wide fishing effort. The linear regression has the following
 344 form:

$$E_{it} = \sum_{t \neq 0} \beta_t T_t + \mu_i + \epsilon_{it} \quad (1)$$

345 where T_t are indicators for periods relative to year of treatment, such that $t = 0$ corresponds
 346 to the last year when purse seine fishing was allowed, 2009. Negative values of t are pre-
 347 closure years, and positive values of t are post-closure years. The parameter of interest is
 348 β_t , which is a vector of coefficients capturing changes in effort inside the high seas pockets
 349 relative to the last year when fishing was allowed (2009). Then, μ_i are grid-cell fixed-effects
 350 and ϵ_{it} is the error term, which accounts for spatio-temporal autocorrelation via Conley
 351 standard errors. These results are shown in Figure B1, which captures changes in effort
 352 relative to 2009. However, our main specification estimates changes across time relative to
 353 changes in effort in other high seas pixels:

$$E_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it}. \quad (2)$$

354 Here, $D_i = 1$ indicates pixel i falls within the high seas closure and $D_i = 0$ indicates other
 355 high seas cells not subject to the closure. τ_t are fixed-effects by year and μ_i is defined as
 356 above. This model allows us to account for region-wide trends in fishing effort, and under
 357 standard assumptions, allows us to derive a causal estimate of the intervention. The results

are shown in Figure 2 Panels C and D.

Aggregate treatment effects were also estimated using the following regression models: $E_{it} = \beta \text{Post}_t + \mu_i + \epsilon_{it}$ and $E_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$. Post_t is a dummy variable that takes the value of 1 or 0 if the observation comes from after or before the implementation period, respectively. Results are shown in Tables 1, B1, and B2.

Note that our pre-registration plan stated that we would also estimate models using log-transformed outcomes. However, due to the large number of zeroes in the data set, we instead use an inverse-hyperbolic sine transformation, considered a standard method for cases like this [49]. Event-study plots are shown in Figures B1 and B2, while aggregate results are shown in Tables B1 and B2.

3.4.2 Testing for changes in CPUE inside the high seas pockets

Let y_{it} denote longline catch-per-unit-effort (CPUE) in pixel i at time t . We estimate:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (3)$$

where all regressors have the same definitions as above. The coefficient of interest, β_t , captures the change in CPUE within the high seas pocket relative to changes observed in other high seas pixels where purse seine effort was not curtailed. A pre-post model with the form $y_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$ was estimated to test for aggregate effects. We also estimated log-linear versions of this model. Results for bigeye tuna are shown in Figure 3 Panels C and D and in Table 2. Results for other specifications and other species are shown in the Supplementary Materials section B.2 in Tables B3–B6 and Figures B3–B6.

3.4.3 Testing for spillover benefits

Let y_{it} denote longline or purse seine catch-per-unit-effort (CPUE) in pixel i at time t . We estimate the following regression:

$$y_{it} = \sum_{t \neq 0} \beta_t (T_t \times D_i) + \mu_i + \tau_t + \epsilon_{it} \quad (4)$$

where the coefficient of interest, β_t , captures the dynamic treatment effect of the high seas closure on CPUE within 100 nautical miles of the high seas pockets, relative to changes observed in other grid cells between 100 and 200 nautical miles from the high seas pockets. A pre-post model with the form $y_{it} = \beta \text{Post}_t \times D_i + \mu_i + \tau_t + \epsilon_{it}$ was also estimated to test for aggregate effects along with log-linear specifications. Results for skipjack tuna in the purse seine fleet are shown in Figure 4 Panels C and D and in Table 3. Results for other specifications and species are shown in Supplementary Materials section B.3 in Tables B7–B14 and Figures B7–B14.

3.4.4 Weights

By construction, $y_{it} = \frac{\sum_{j \in (i,t)} \text{Catch}_j}{\sum_{j \in (i,t)} \text{Effort}_j}$, so $\text{Var}(y_{it}) \propto \frac{1}{\text{Effort}_{it}}$. We therefore assume

$$\text{Var}(\epsilon_{it} \mid X) = \frac{\sigma^2}{\omega_{it}}, \quad \omega_{it} = \text{Effort}_{it},$$

and estimate all CPUE models via weighted least squares using effort as weights.

3.4.5 Software and Reproducibility

All data cleaning and sample construction was performed with the tidyverse (v2.0.0), sf (1.0-21), and terra (1.8-60) packages using R (v4.5.1) and RStudio (2026.07.1 Build 147 “Pacific Dogwood” [50, 51, 52, 53, 54]). All models were fit using the fixest package (v0.13.2; [55]). Regression tables were built using the modelsummary package (v1.3.0; [56]), and event-study plots were built with ggfixest (v0.4.0, [57]) and customized with the ggplot2 package (v4.0.2, [58]). All our data and code are publicly available in a GitHub repository posted at https://github.com/jcvdav/PNA_HS_closure.

Acknowledgments

J.C.V.D. received funding from the PEW charitable trust. The funders had no say in the design or execution of the research.

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Appendix

Supplementary Materials

For

**Limited Conservation Benefits From the First Marine Protected
Area in the High Seas**

A Supplementary Figures and Tables

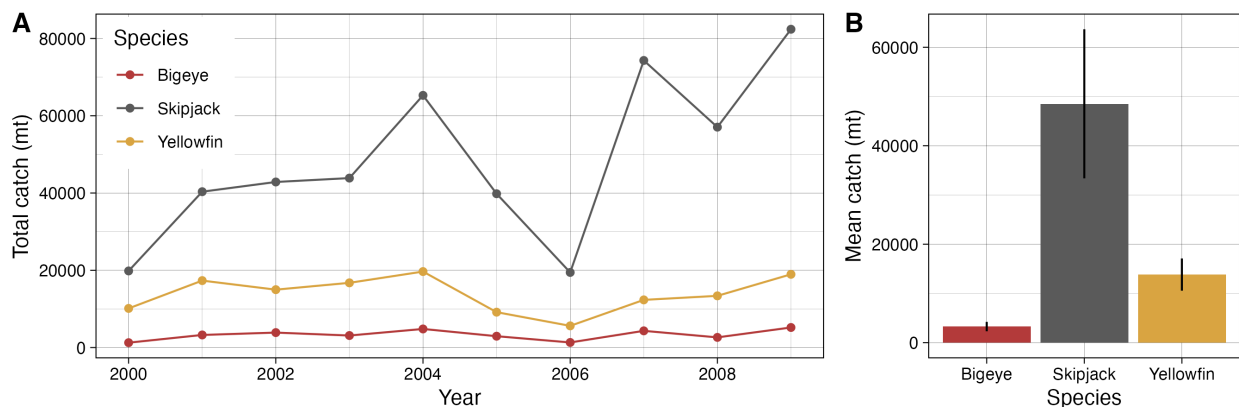


Figure A1: Historical (2000-2009) catch of bigeye, skipjack, and yellowfin tuna by tuna purse seiners fishing in the high seas pockets before their closure. Panel A shows a time series of catch by species, panel B shows mean ($\pm$ SD) catch by species.

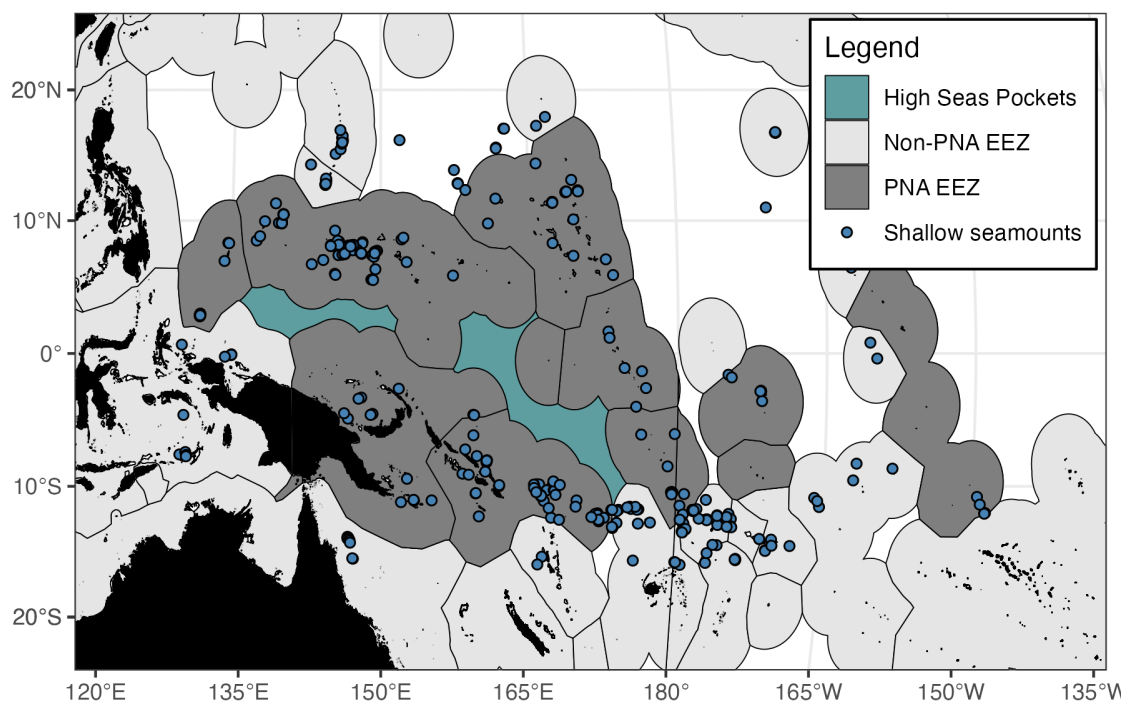


Figure A2: Map of shallow seamounts in the Western and Central Pacific. Blue points indicate the centroid of shallow seamounts (≤ 150 m deep) as defined in [44, 45]. Light gray polygons show the Exclusive Economic Zones of all nations. Dark gray polygons highlight the Exclusive Economic Zones of PNA nations and Tokelau. The two teal polygons show the High Seas Pockets closed to purse seine fishing starting in 2010.

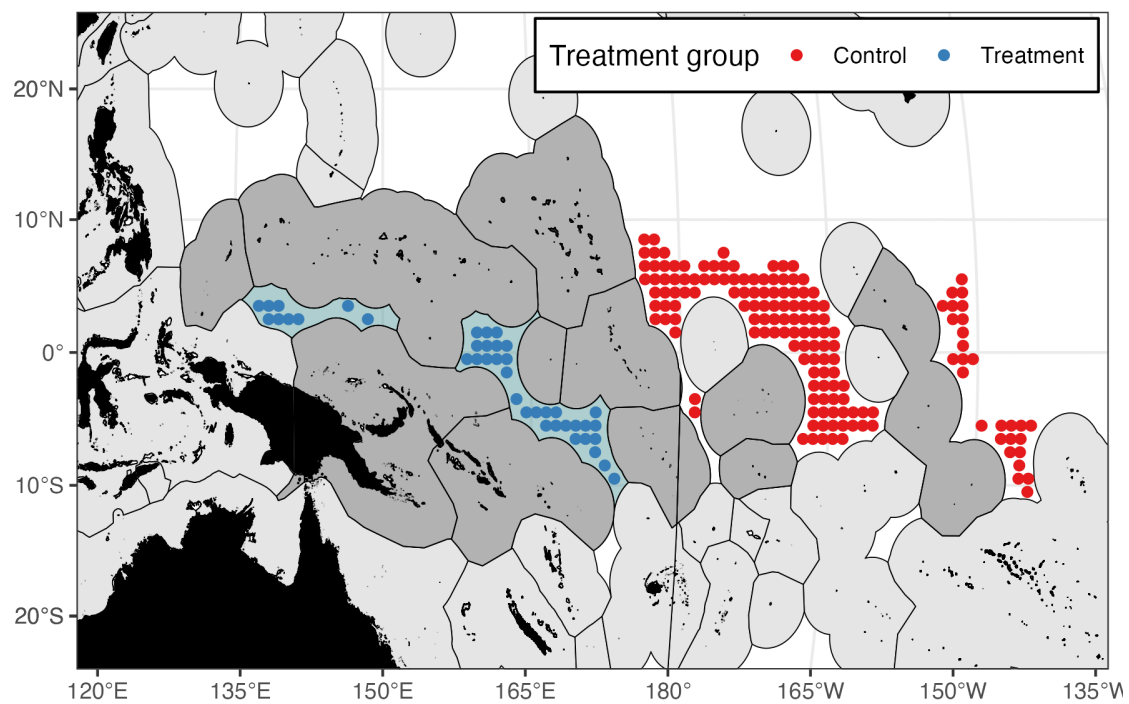


Figure A3: Map of location of treatment and control cells used in our test of changes in purse seine fishing effort in the high seas pockets.

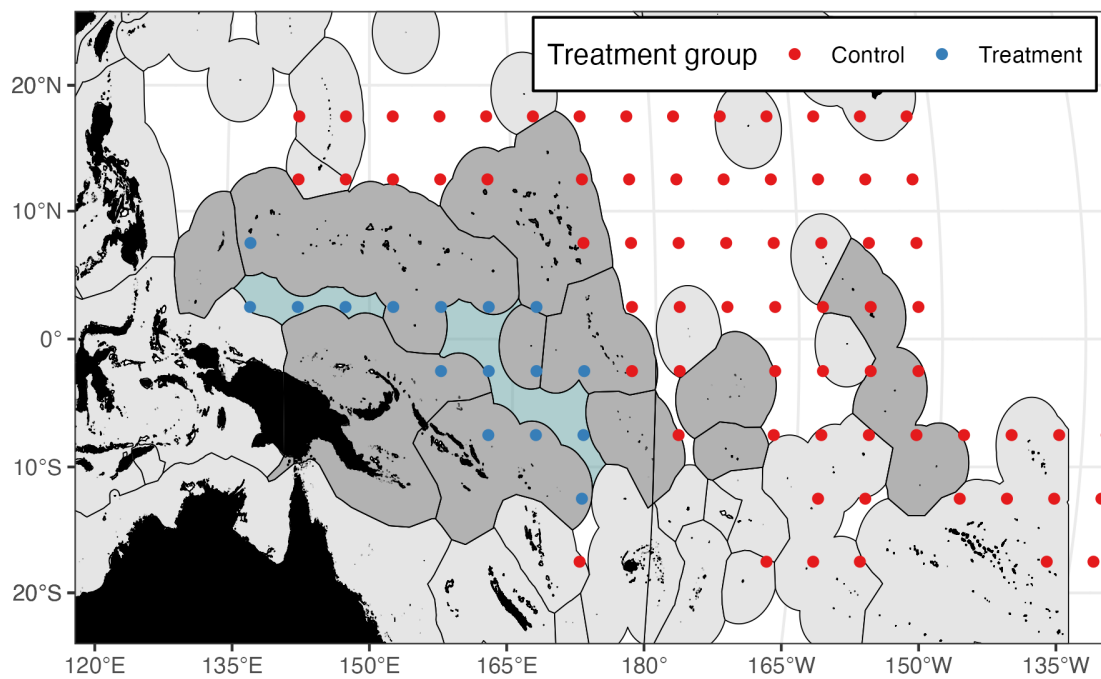


Figure A4: Map of location of treatment and control grid cells used in our test of changes in CPUE for the longline fleet operating in the high seas pockets.

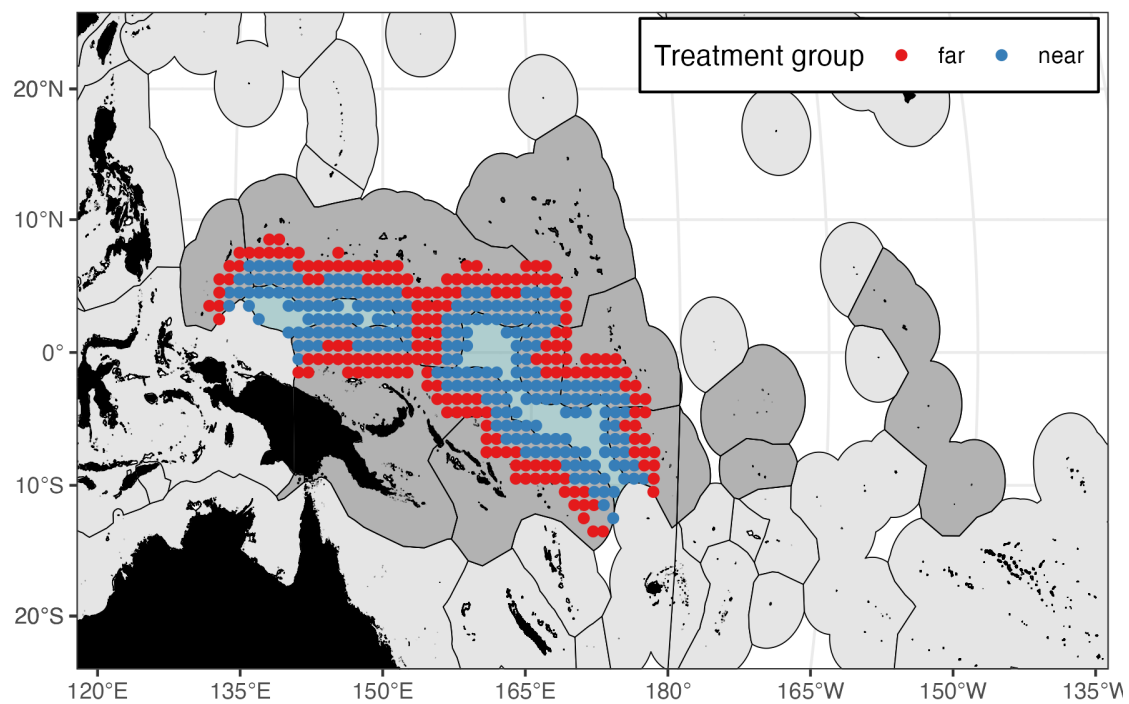


Figure A5: Map of location of treatment and control grid cells used in our test of a spillover benefit to the purse seine fleet.

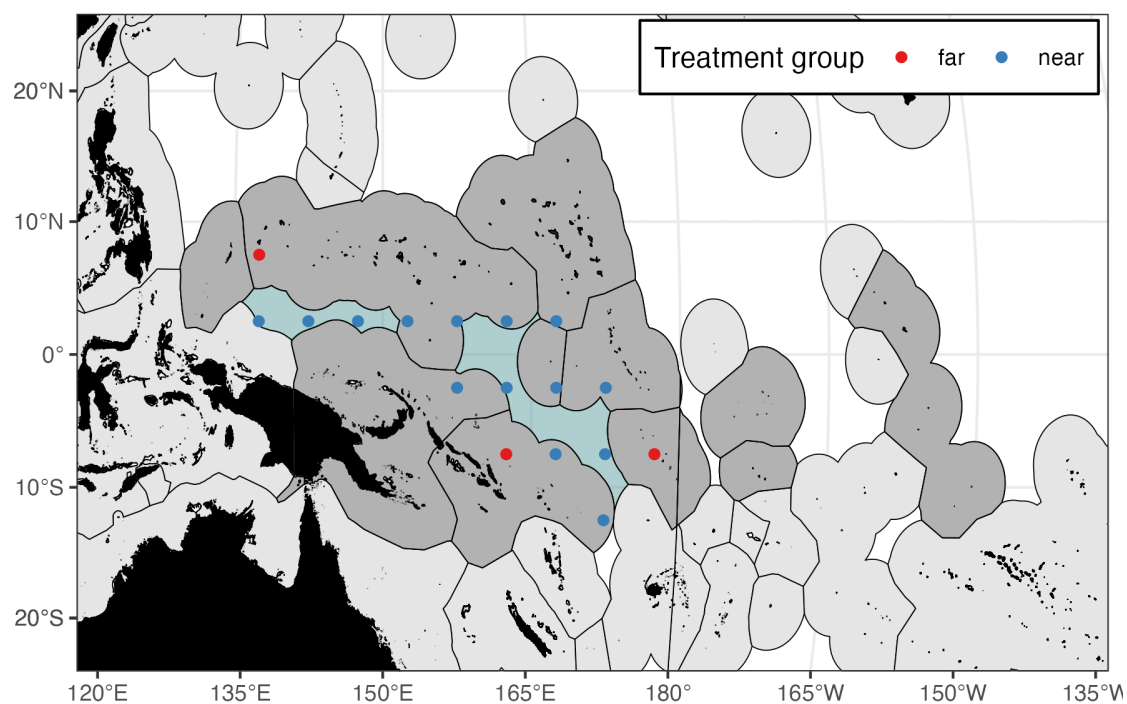


Figure A6: Map of location of treatment and control grid cells used in our test of a spillover benefit to the longline fleet.

B Additional Results

B.1 Decrease in purse seine effort in high seas pockets

The reduction in purse seine fishing effort in the high seas pockets is robust across model specifications and outcome transformations. A within-treated specification, which estimates changes relative to the 2009 baseline without a control comparison, shows significant reductions of 54.55 days and 44.02 sets per grid cell per year ($p < 0.01$; Table B1), with event-study estimates confirming no pre-treatment trends (Figure B1). Results comparing trends with other high seas areas within the WCPFC convention area also remain significant when outcomes are transformed using the inverse hyperbolic sine (Table B2; Figure B2).

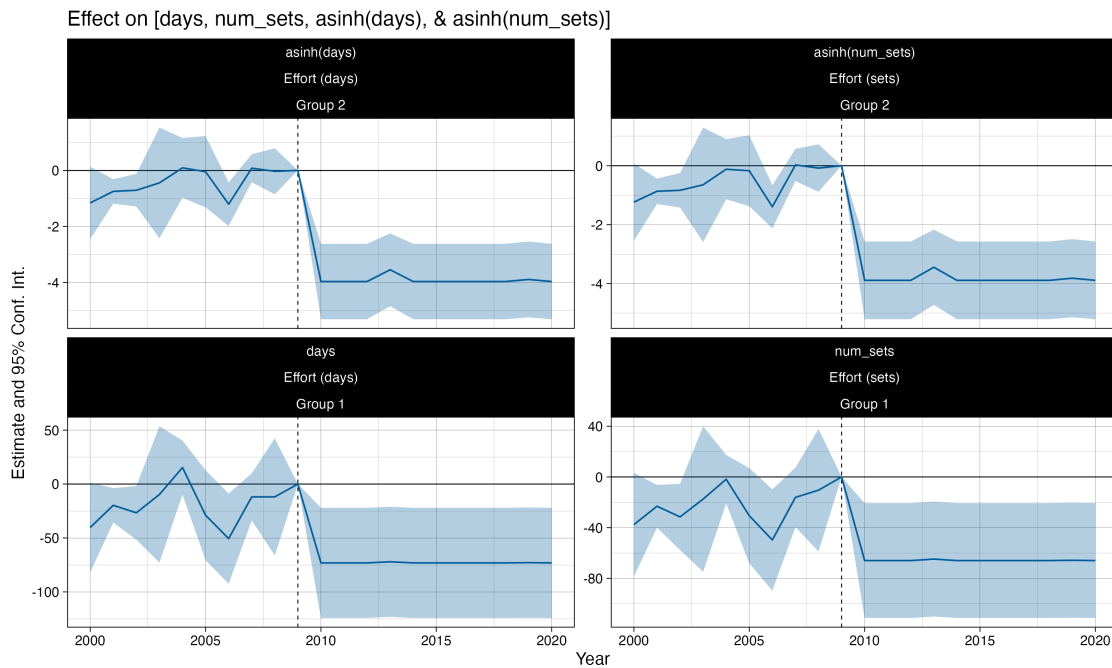


Figure B1: Event-study estimates of the change in purse seine fishing effort within the high seas pockets, estimated using only treated grid cells relative to the 2009 baseline (without a control comparison). Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels).

Table B1: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure. Panel A presents results in levels. Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

	Effort (days)	Effort (sets)
<i>A) Levels</i>		
Post	-54.555*** (13.639)	-44.018*** (10.394)
R2 Adj.	0.363	0.332
<i>B) Inverse-hyperbolic sine transformation</i>		
Post	-3.505*** (0.452)	-3.310*** (0.424)
R2 Adj.	0.675	0.662
Num.Obs.	840	840

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. Numbers in parentheses are Conley standard errors using a 220km radius.

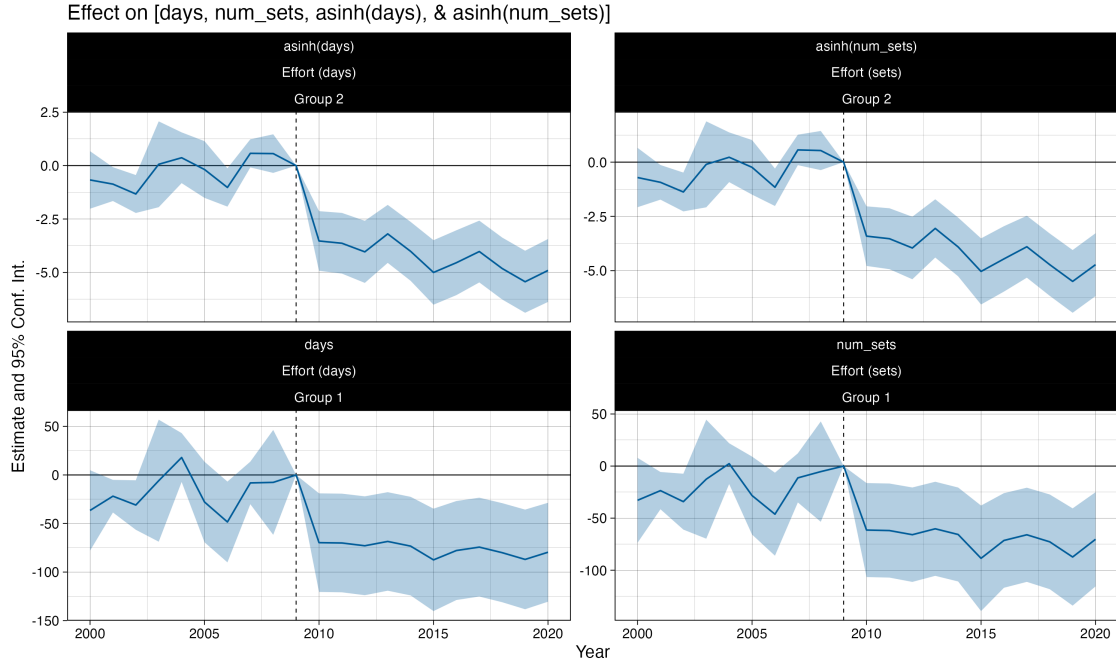


Figure B2: Event-study estimates of the change in purse seine fishing effort in the high seas pockets, relative to fishing effort in comparable high seas regions in the WCPFC convention area. Estimates are shown for fishing effort measured in days (left panels) and number of sets (right panels), under the inverse hyperbolic sine (IHS) transformation (top panels) and in levels (bottom panels). The bottom panels are equivalent to the main text results shown in Figure 2.

Table B2: Coefficient estimates for change in fishing effort inside the high seas pockets after the closure, relative to changes in fishing effort observed in other comparable high seas areas of the WCPFC convention area. Panel A presents results in levels (identical to the main-text estimates in Table 1). Panel B presents results in which the dependent variable is transformed using the inverse hyperbolic sine (IHS) transformation.

	Effort (days)	Effort (sets)
<i>A) Levels</i>		
Post x Treated	-59.470*** (13.762)	-50.933*** (10.607)
R2 Adj.	0.388	0.301
<i>B) Inverse-hyperbolic sine transformation</i>		
Post x Treated	-4.036*** (0.478)	-3.881*** (0.451)
R2 Adj.	0.465	0.451
Num.Obs.	4011	4011

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors using a 220km radius.

B.2 Change in longline CPUE in high seas pockets

Here we present results for changes in longline CPUE within the high seas pockets for all species and additional model specifications. Across all species combined (Table B3; Figure B3) and for albacore tuna (Table B4; Figure B4), we detect no significant change in longline CPUE. The bigeye tuna results shown in the main text are qualitatively similar under log-transformation, though we note a marginally significant decline in CPUE when measured in metric tons (Table B5; Figure B5). For yellowfin tuna, we estimate a significant increase in log-transformed CPUE measured in number of fish ($p < 0.05$; Table B6; Figure B6).

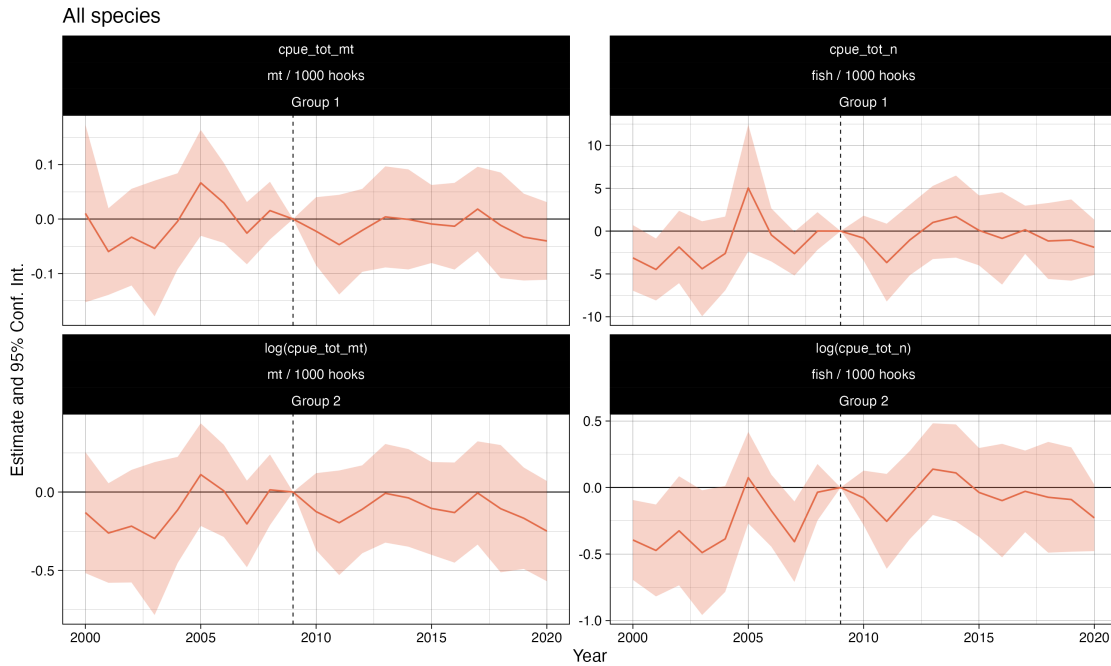


Figure B3: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B3: Coefficient estimates for change in all tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	fish / 1000 hooks	mt / 1000 hooks
<i>A) Levels</i>		
Post x Treated	0.664 (1.605)	-0.009 (0.028)
R2 Adj.	0.637	0.507
<i>B) Log-transformed</i>		
Post x Treated	0.187 (0.139)	-0.003 (0.089)
R2 Adj.	0.700	0.616
Num.Obs.	1632	1632

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

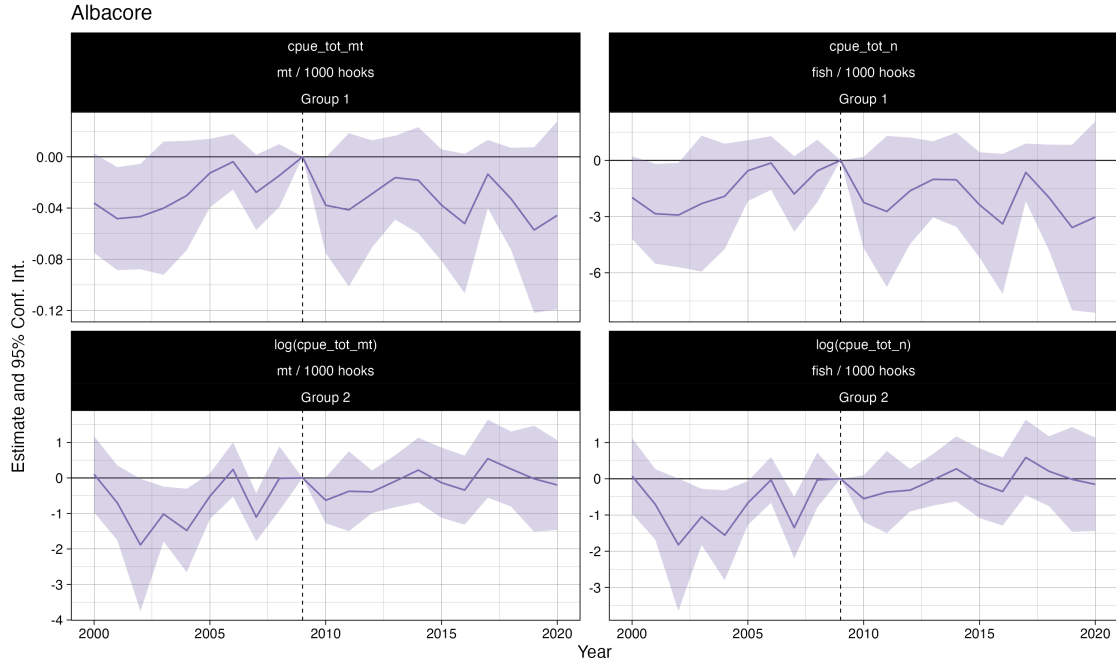


Figure B4: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B4: Coefficient estimates for change in albacore tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	fish / 1000 hooks	mt / 1000 hooks
<i>A) Levels</i>		
Post x Treated	-0.645 (0.793)	-0.009 (0.012)
Num.Obs.	1606	1606
R2 Adj.	0.856	0.852
<i>B) Log-transformed</i>		
Post x Treated	0.644 (0.498)	0.540 (0.485)
Num.Obs.	1606	1603
R2 Adj.	0.821	0.809

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

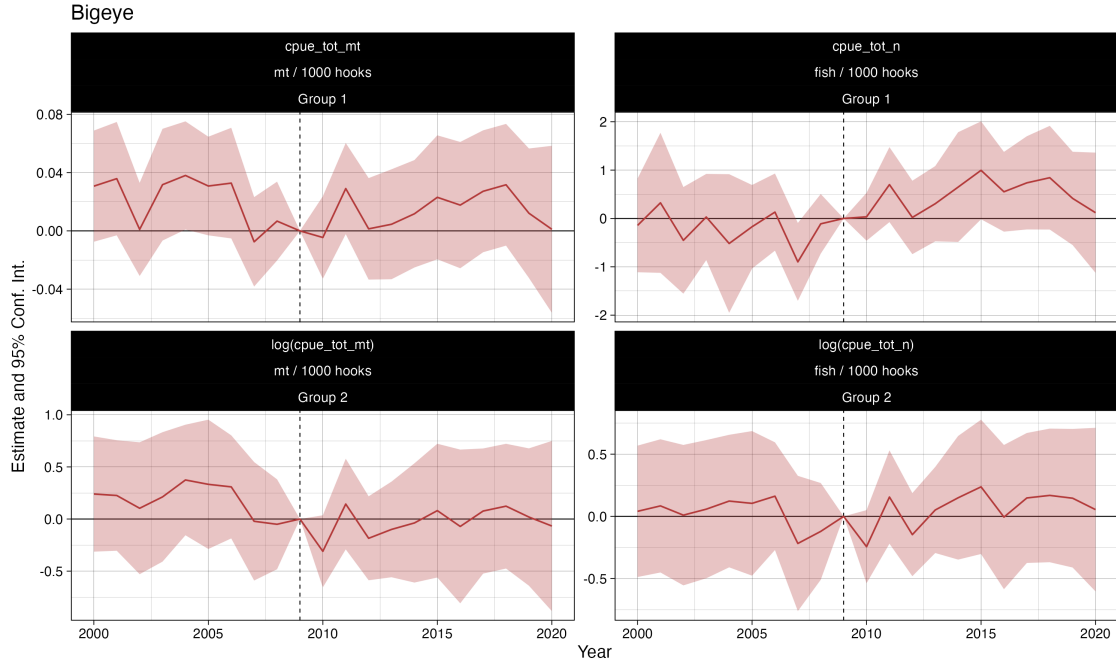


Figure B5: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 3.

Table B5: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels (same as in Table 2). Panel B presents results in which the dependent variable is log-transformed.

	fish / 1000 hooks	mt / 1000 hooks
<i>A) Levels</i>		
Post x Treated	0.697* (0.363)	-0.003 (0.011)
R2 Adj.	0.798	0.829
<i>B) Log-transformed</i>		
Post x Treated	0.055 (0.106)	-0.176* (0.094)
R2 Adj.	0.852	0.880
Num.Obs.	1631	1631

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

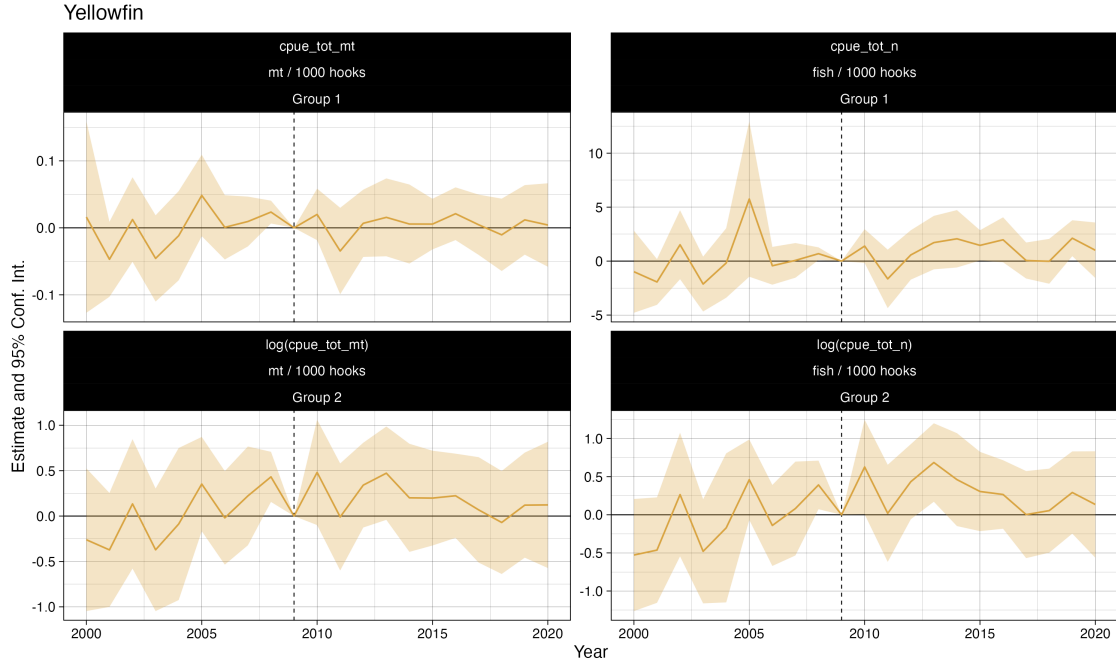


Figure B6: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna in the high seas pockets, relative to CPUE in other tropical (20°S to 20°N) high seas regions in the WCPFC convention area. Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B6: Coefficient estimates for change in yellowfin tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	fish / 1000 hooks	mt / 1000 hooks
<i>A) Levels</i>		
Post x Treated	0.598 (0.857)	0.002 (0.016)
R2 Adj.	0.538	0.607
<i>B) Log-transformed</i>		
Post x Treated	0.329** (0.151)	0.171 (0.117)
R2 Adj.	0.702	0.704
Num.Obs.	1632	1632

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

B.3 Spillover benefits to fisheries adjacent to high seas pockets

Purse seine. We find no evidence of spillover benefits to the purse seine fleet operating near the high seas pockets for any species or model specification. Results are null across all species combined (Table B7; Figure B7), bigeye tuna (Table B8; Figure B8), skipjack tuna (Table B9; Figure B9), and yellowfin tuna (Table B10; Figure B10), in both levels and log-transformed specifications.

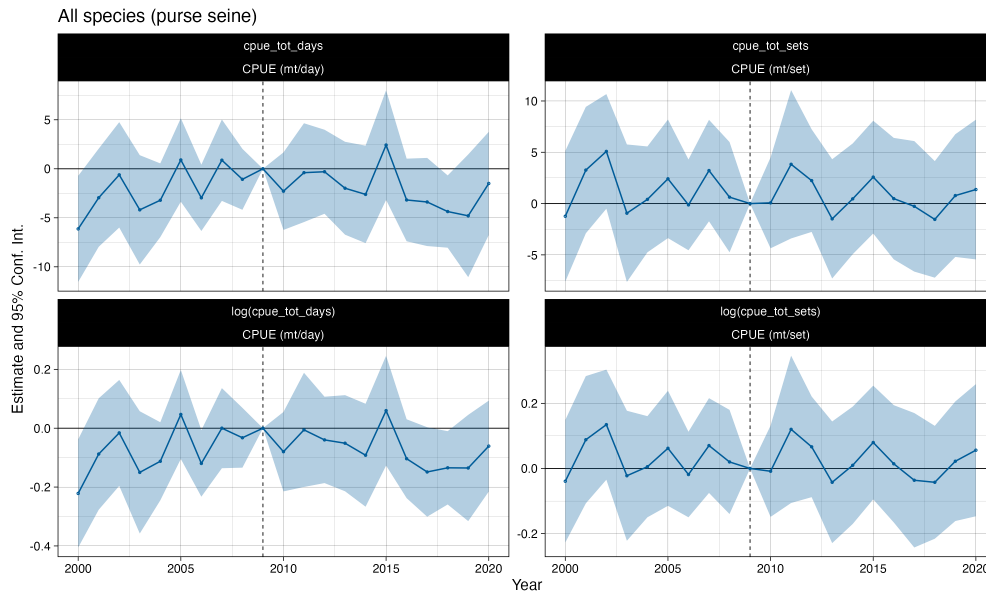


Figure B7: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B7: Coefficient estimates for change in all tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (mt/day)	CPUE (mt/set)
<i>A) Levels</i>		
Post x Near	-0.294 (0.827)	-0.468 (1.050)
R2 Adj.	0.273	0.209
<i>B) Log-transformed</i>		
Post x Near	-0.007 (0.028)	-0.007 (0.034)
R2 Adj.	0.258	0.221
Num.Obs.	5960	5960

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

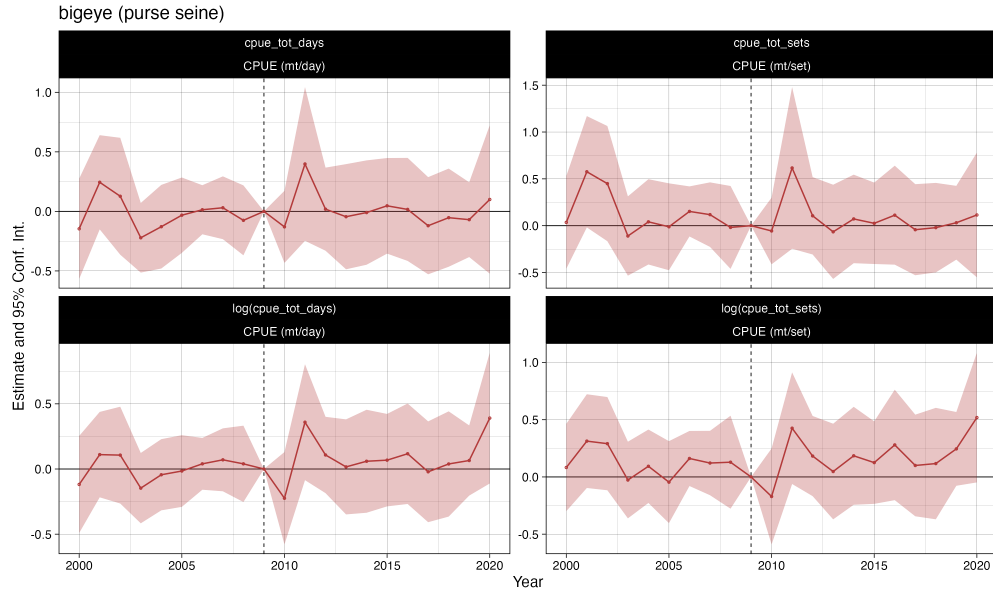


Figure B8: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B8: Coefficient estimates for change in big-eye tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (mt/day)	CPUE (mt/set)
<i>A) Levels</i>		
Post x Near	0.055 (0.104)	-0.005 (0.121)
R2 Adj.	0.174	0.209
<i>B) Log-transformed</i>		
Post x Near	0.086 (0.086)	0.081 (0.096)
R2 Adj.	0.227	0.279
Num.Obs.	5908	5908

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

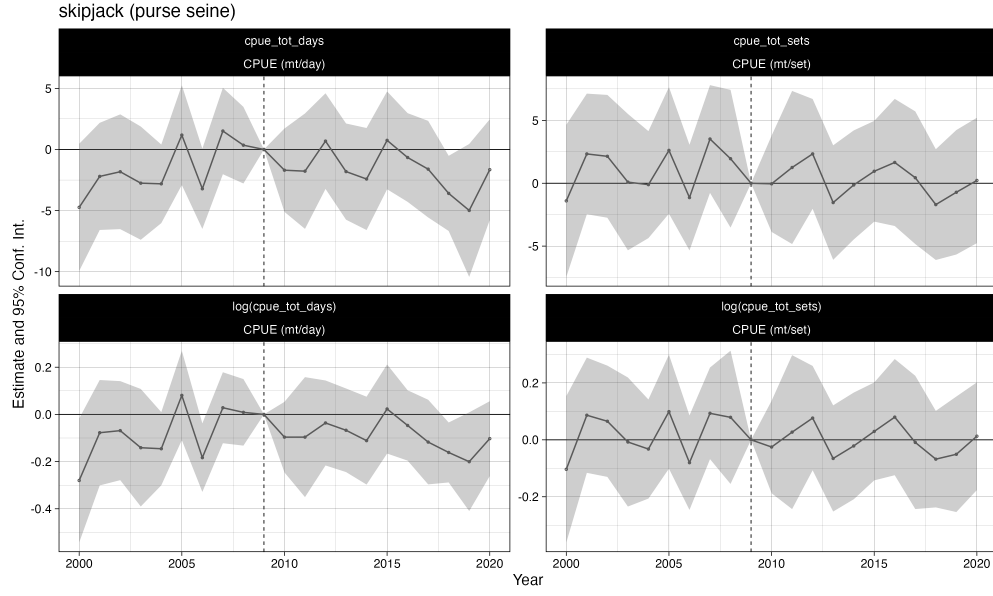


Figure B9: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for skipjack tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels). The top panels are equivalent to the main text results shown in Figure 4.

Table B9: Coefficient estimates for change in skip-jack tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (mt/day)	CPUE (mt/set)
<i>A) Levels</i>		
Post x Near	-0.496 (0.679)	-0.865 (0.818)
R2 Adj.	0.333	0.245
<i>B) Log-transformed</i>		
Post x Near	-0.024 (0.031)	-0.026 (0.038)
R2 Adj.	0.316	0.244
Num.Obs.	5960	5960

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

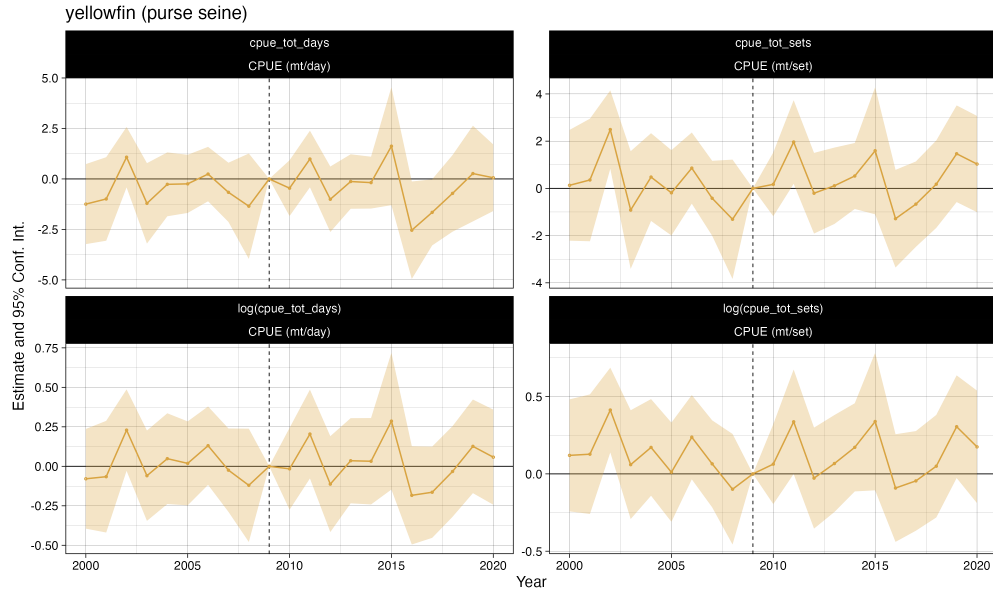


Figure B10: Event-study estimates of the change in purse seine catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Both the near (treated) and far (control) samples are restricted to grid cells within PNA member Exclusive Economic Zones. Estimates are shown for CPUE measured in tons per day (left panels) and tons per set (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B10: Coefficient estimates for change in yellowfin tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles and inside PNA nation's Exclusive Economic Zones. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (mt/day)	CPUE (mt/set)
<i>A) Levels</i>		
Post x Near	0.146 (0.281)	0.402 (0.300)
R2 Adj.	0.221	0.280
<i>B) Log-transformed</i>		
Post x Near	0.015 (0.037)	0.022 (0.044)
R2 Adj.	0.244	0.304
Num.Obs.	5956	5956

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

634 **Longline.** There are only 3 cells within 100–200 nautical miles, leaving us with a reduced
635 control group for this portion of the analyses. These results are presented to comply with
636 our preregistration, but we note the lack of robustness that comes with such a limited
637 sample size. With this caveat in mind, we detect no significant spillover effects for all species
638 combined (Table B11; Figure B11), bigeye tuna (Table B13; Figure B13), or yellowfin tuna
639 (Table B14; Figure B14). The only marginally significant result is for albacore tuna under
640 the log-transformed specification ($p < 0.1$; Table B12; Figure B12).



Figure B11: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) across all species combined within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B11: Coefficient estimates for change in all tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (fish/1000 hooks)	CPUE (mt/1000 hooks)
<i>A) Levels</i>		
Post x Near	2.983 (2.918)	0.015 (0.039)
R2 Adj.	0.578	0.360
<i>B) Log-transformed</i>		
Post x Near	0.145 (0.174)	0.057 (0.128)
R2 Adj.	0.662	0.404
Num.Obs.	347	347

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

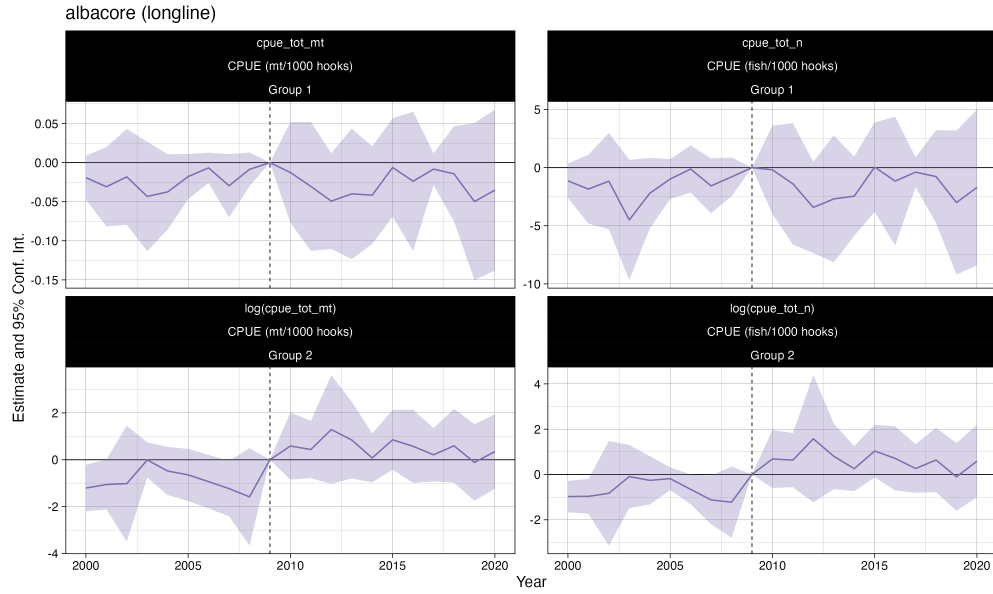


Figure B12: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for albacore tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B12: Coefficient estimates for change in albacore tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (fish/1000 hooks)	CPUE (mt/1000 hooks)
<i>A) Levels</i>		
Post x Near	-0.487 (1.260)	-0.011 (0.019)
R2 Adj.	0.878	0.881
<i>B) Log-transformed</i>		
Post x Near	1.326* (0.724)	1.403* (0.810)
R2 Adj.	0.843	0.819
Num.Obs.	339	339

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

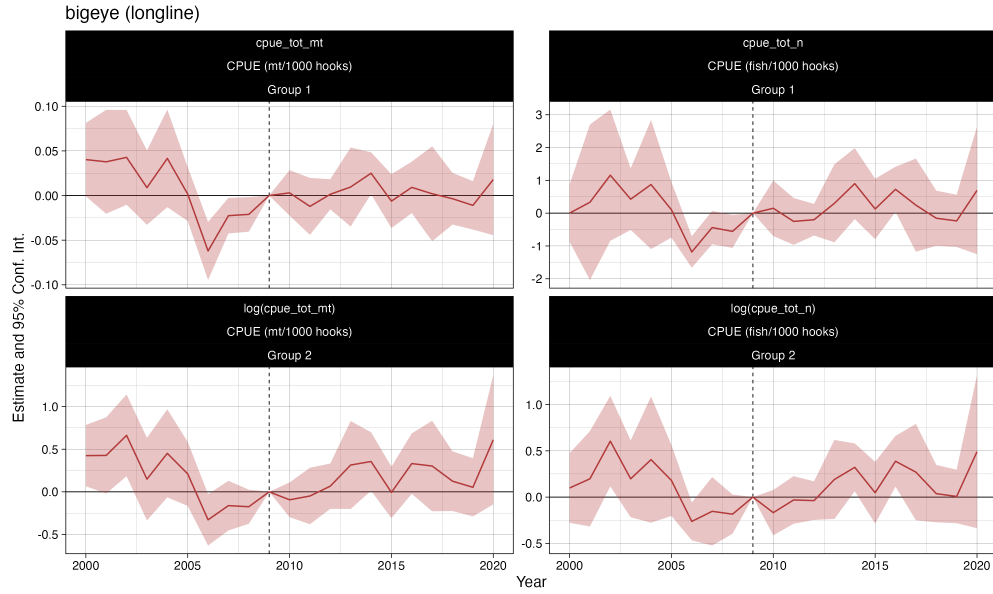


Figure B13: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for bigeye tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

Table B13: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (fish/1000 hooks)	CPUE (mt/1000 hooks)
<i>A) Levels</i>		
Post x Near	0.136 (0.433)	0.000 (0.013)
R2 Adj.	0.683	0.709
<i>B) Log-transformed</i>		
Post x Near	0.028 (0.134)	0.032 (0.152)
R2 Adj.	0.812	0.848
Num.Obs.	347	347

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

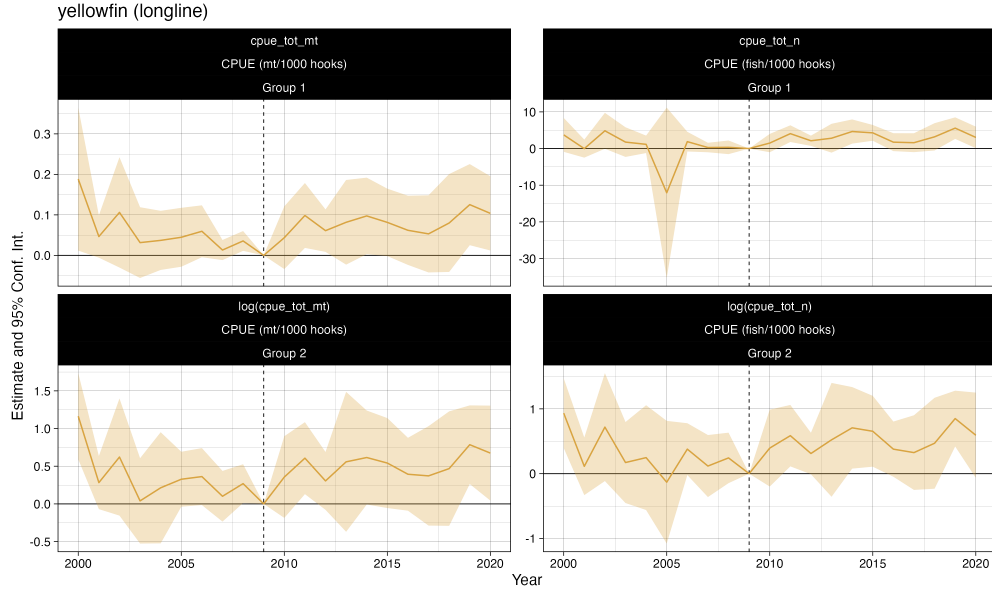


Figure B14: Event-study estimates of the change in longline catch-per-unit-effort (CPUE) for yellowfin tuna within 100 nm of the high seas pockets, relative to changes in CPUE observed in areas further away (100–200 nm). Estimates are shown for CPUE measured in metric tons per 1000 hooks (left panels) and number of fish per 1000 hooks (right panels), in levels (top panels) and for log-transformed CPUE (bottom panels).

C Robustness Tests

C.1 Results by high seas pocket

To assess whether our main results are driven by a single pocket, and as a robustness test to account for the exemption granted to the Philippines, we re-estimate the main specifications for bigeye tuna longline CPUE (Table 2) and skipjack tuna purse seine CPUE spillover (Table 3) separately for each high seas pocket. For the longline CPUE analysis, we drop treated grid cells from one pocket at a time while retaining the full set of control cells. For the spillover analysis, we split the entire sample—both near and far grid cells—east and west of 152.5°. The increase in bigeye CPUE measured in number of fish is significant for both the western ($p < 0.01$) and eastern ($p < 0.05$) pockets, with a larger point estimate in the western pocket (Table C1). The null spillover result for skipjack tuna holds for both pockets

Table B14: Coefficient estimates for change in yellowfin tuna CPUE caught by the longline fleet in areas within 100 nautical miles of the high seas pockets after the closure, relative to changes in CPUE observed for areas between 100 and 200 nautical miles. Panel A presents results in levels. Panel B presents results in which the dependent variable is log-transformed.

	CPUE (fish/1000 hooks)	CPUE (mt/1000 hooks)
<i>A) Levels</i>		
Post x Near	3.346 (2.147)	0.026 (0.030)
R2 Adj.	0.383	0.382
<i>B) Log-transformed</i>		
Post x Near	0.254 (0.209)	0.176 (0.200)
R2 Adj.	0.518	0.493
Num.Obs.	347	347

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.

Table C1: Coefficient estimates for change in bigeye tuna CPUE caught by the longline fleet in the high seas pockets after the closure, relative to changes in CPUE observed for other tropical (20°S - 20°N) high seas areas in the WCPFC convention area. Models are estimated separately for HSP1 (western pocket, $\text{lon} \leq 152.5^\circ$) and HSP2 (eastern pocket, $\text{lon} > 152.5^\circ$; See Figure 1). The first column shows our main text estimates as in Table 2.

	sample: Full sample	sample: HSP1	sample: HSP2
<i>A) CPUE (fish / 1000 hooks)</i>			
Post x Treated	0.697* (0.363)	1.537*** (0.520)	0.397** (0.196)
R2 Adj.	0.798	0.801	0.804
<i>B) CPUE (mt / 1000 hooks)</i>			
Post x Treated	-0.003 (0.011)	0.020 (0.013)	-0.011 (0.010)
R2 Adj.	0.829	0.830	0.835
Num.Obs.	1631	1407	1529

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 1100km radius.

652 individually (Table C2).

Table C2: Coefficient estimates for change in skipjack tuna CPUE caught by the purse seine fleet in areas within 100 nautical miles of each high seas pocket after the closure. Models are estimated separately for HSP1 (western pocket, $\text{lon} \leq 152.5^\circ$) and HSP2 (eastern pocket, $\text{lon} > 152.5^\circ$); See Figure 1. The first column shows our main text estimates as in Table 3.

	sample: Full sample	sample: HSP1	sample: HSP2
<i>A) CPUE (mt/day)</i>			
Post x Near	-0.496 (0.679)	-0.132 (0.635)	-0.952 (0.957)
R2 Adj.	0.333	0.433	0.319
<i>B) CPUE (mt/set)</i>			
Post x Near	-0.865 (0.818)	-0.925 (0.715)	-1.206 (1.141)
R2 Adj.	0.245	0.353	0.213
Num.Obs.	5960	1645	4315

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

The unit of observation is a grid cell in a given year. All model specifications include fixed effects by year and grid cell. Numbers in parentheses are Conley standard errors with a 220km radius.